

POAMS: LECTURE 3

PHILOSOPHICAL BACKGROUND: NORMAL REALISM

*Having covered some of the history of Normal Realism and POAMS, in this lecture I provide some of the philosophical underpinning to POAMS, leading to the Normal Realist philosophy. In this approach, as in Relativity, the role of the **observer** is key. The lecture is based to some extent on the material in Chapter 1 of 'Light-Speed....'*

In order to understand how Normal Realism came about, we need to study some of the development of philosophical ideas over the past 350 years.

1. God's-eye-view Realism

One of the founders of what we now know as Physics was the French philosopher, René Descartes (1596-1650). He argued that all we ever see or sense are our 'ideas' of physical objects, never those objects themselves as 'God' may be presumed to see them. For Descartes this was no problem since, as he reasoned, the God who made those objects also made the senses by which we perceive them; and since 'God is no deceiver', it followed that what our senses tell us must be true. Thus, Descartes concluded that what we perceive is an exact replica of, and hence as real as, what God sees. This presumed absolutist 'God's-eye-view' of things is the basis of what has since become known, in modern Physics, as *Realism*.

Later philosophers, not surprisingly, saw Descartes' argument as fallacious. As his critics saw it, Descartes' argument splits our minds in two. In the one part are our ideas of what things are 'in themselves', as we presume 'God' sees them, and in the other are our ideas of what those things are as 'we' see them. And then we have yet another set of ideas as to how these two separated sets of ideas in our minds are connected, which Descartes conceived as a linkage provided by a non-deceitful God.

This view of Descartes, this splitting-off of a presumed 'God's-eye-view' of physical objects from our perceptions of them, created a false dichotomy. Called *Cartesian dualism*, this split what was traditionally known as Natural Philosophy in two. The 'God's-eye-view' half became Physics and the other half, the 'Man's-eye-view', became Psychology. This dichotomy between 'Matter' and 'Mind', introduced by Descartes, has held sway in Western thinking right up to the present time.

2. Subjective Idealism

With this false dichotomy of Matter and Mind, Descartes had inadvertently set a trap for future philosophers. Among the first to fall into this trap was the highly influential philosopher, Bishop George Berkley (1685-1753). Berkley reasoned that since we cannot presume to be ‘God’, we can have no knowledge as to how God might perceive things. There was no point, therefore, in this splitting of our idea-systems into, on the one hand, our ideas of what things are ‘as we perceive them’ and, on the other hand, as *God* perceives them’. Hence, Berkley concluded that everything that *is* exists ‘*in the mind of the observer*’.

Some philosophers thought that Berkley’s was a good logical solution to Cartesian dualism, although it simply compounded the fallacy. Among these were the prominent writers, David Hume (1711-1776) and Immanuel Kant (1724-1804). These philosophers, known as the *English Empiricists*, in their efforts to establish an alternative to Cartesian dualism, accepted without question the very dichotomy they were trying to remove. It left them with a completely truncated view of knowledge in which *nothing exists beyond perception*. In this way was created the longstanding, ‘Problem of Knowledge’, which was the completely bogus question of how Mind, the subject of Philosophy and Psychology, can ever be certain even of the existence of Matter, the subject of Physics.

In this way, Berkley’s philosophy became the basis of a radical *relativism*, subsequently known as *phenomenalism*, which many mistakenly thought to be a theory of reality based *entirely* on appearances. This was the view known as *Subjective Idealism*. The infamous dictum which launched this subjectivist view was Berkley’s statement that ‘*esse is percipi*’ (to be is to be perceived). In its logical extreme, this becomes the calamitous view called *Solipsism*, the notion that the ‘Mind’ in which everything exists is, ultimately, one’s own mind, so that there can be no objective distinction between dream and reality. From Berkley’s time on, it could be argued that a lot of the energy that went into the pursuit of academic Philosophy was squandered in pursuing vainly contrived ways of escaping this solipsistic consequence.

Nevertheless, during the 19th century, the phenomenalism developed by Kant and others was articulated as a basis for modern Relativistic Physics by Mach. For Mach, all our knowledge of the world (especially in science)

is based on *sense-impressions*, which he called ‘sense-data’. In consequence, the emphasis was placed on **observation**, as opposed to theoretical speculation. Such was the influence of Mach’s writings that in Physics, from then on, the primacy of physical observations began to be emphasised.

3. The rise and fall of the ‘Linguistic’ movement

At the beginning of the 20th century, a Cambridge philosopher, George E. Moore (1873-1958) subjected Berkley’s infamous dictum to a method which was to become known as *Linguistic Analysis*. Moore argued that Berkley’s statement cannot be true by definition, nor is it factually true. In what sense, then, can it possible be true that in order to *be* or *exist*, an object has to be seen or perceived? The conclusion of Moore’s analysis was that there is simply no warrant for Berkley’s statement; that it is a form of language which has no true meaning. Moore’s analysis was summarised in his essay ‘The Refutation of Idealism’, published in *Mind* in 1903. This one essay quietly changed the course of Western philosophy.

Following Moore, other writers began to seek to purge the common language altogether of what they perceived as other relics of Cartesian dualism. These writers included the eminent philosophers, John L. Austin (1911-1960), Gilbert Ryle (1900-1976) and Ludwig Wittgenstein (1889-1951). In this way, the ‘Linguistic’ movement in Philosophy flourished. The ultimate aim of these Linguistic philosophers was, by applying Linguistic Analysis, to purify and restore to ‘normality’ a contaminated common language. This they aimed to achieve by an insistence on the ‘proper usage’ of what they thought of as ‘Ordinary Language’, which was inevitably the ‘King’s English’ of these Oxbridge scholars.

The ‘Linguistic’ movement came to the conclusion that, as far as human knowledge is concerned, our *ideas* of what a thing is and what that thing is in itself, ‘behind those ideas’, are never two things but one. So, what had been classically conceived as ‘The Problem of Knowledge’ of fitting our *ideas* of physical reality to that physical reality itself had never been anything more than the problem of fitting one set of ideas to another, which is a different problem entirely from what had been imagined since the time of Descartes.

For example, as Austin would say, the only difference between a *real* barn door and an *idea* of that barn door is between two different sets of observational impressions. The one involves, say, walking up to that object in daylight, knocking on it, picking at it with a penknife, checking that other observers concur with our findings and so on. The other consists of, say, thinking about that object, having some vague impression of it whilst feeling about in the dark, seeing it at a distance in a dim light, or in some other similarly disadvantaged way. Since we can never perceive the barn door as ‘God’ might be presumed to perceive it behind *all* those various impressions, no appeal to any such object, as the datum of ‘reality’ can add anything whatsoever to what we already know about it with the best of our available sense perceptions, both direct and instrumental.

However, in their zeal to apply this new analysis to language as a whole, the ‘Linguistic’ movement committed yet another fallacy. It was made to seem as though, in Wittgenstein’s memorable words, ‘The World is Language’. For practically minded scientists, this ‘Linguistic’ philosophy was no advance on Subjective Idealism. For them, it was too much to think of what is revealed to us by microscopes, telescopes, voltmeters and so on, as simply ‘bits of language’. In consequence, this led to a final parting of the ways between modern ‘Realistic’ Science and ‘Idealistic’ Philosophy.

The highpoint of this conflict between Science and Philosophy came in the 1960s and was due to the indiscriminate dismissal by the ‘Linguistic’ philosophers of all objective modern scientific language as incurably polluted by the jargon created initially by Cartesian dualism. This jargonised language they disparagingly dubbed ‘Scientism’. The trouble was that, in consequence, these philosophers rejected scientific language *as a whole*.

What these ‘Linguistic’ philosophers had argued, however, was not entirely false. Clearly, in our commonsense and scientific observations of the world, nothing comes labelled by nature as, for example, an ‘electron’, a ‘quark’, or a ‘quasar’. It is *we* who do that labelling or naming, and it is we who arbitrarily assign these objects to their various language classes and categories. This defines the normal creative, ongoing process of concept-construction and revision generally known as *language*.

The mistake that the ‘Linguistic’ philosophers made was to assume that there is some ‘pure’ form of language which, by virtue of nothing but its

own formality, constitutes an eternal and ‘incurable’ template into which all the scientifically discoverable things and processes automatically fit. Then apart from its most practical uses in advancing technology, this philosophy would make all Science as such redundant.

But of course, there is no way in which our finite *human* language may, by its own internal efforts, guarantee absolute certainty in that imagined way. This is why ‘Linguistic’ philosophy eventually died a death, taking with it, in many instances, the very name ‘Philosophy’ with which it had become too closely identified. Since then, the sciences have been left to go their various eclectic ways, practically bereft of any awareness of the insecurity of their own unspoken and un-analysed philosophical underpinnings.

In our philosophical approach, we acknowledge the plain fact that *human* language is never self-sufficient as a means of assigning absolute truth and reality to our perceptions of things; that at every stage of its evolution its efficacy has to be judged with reference to something which is super-lingual but not divisive in the Cartesian way. This brings us to ‘Normal Realism’.

4. Normal Realism

Normal Realism is the paradigm of Natural Philosophy which is the logical result of removing from contemporary Physics all vestiges of both the Cartesian and the Berkeleyan fallacies. Based on the phenomenalism of Mach, it exorcises, along the lines laid down by Moore, the spectre of Subjective Idealism that has persisted in phenomenalism until quite recently, and retains some elements of the later ‘Linguistic’ philosophy. The word ‘normal’ in Normal Realism was carefully chosen at its inception in the 1970s, to be synonymous with *normative* or *prescriptive*, in the classical manner of the pre-Cartesian Natural Philosophy from which, originally all physical science sprang. In short, Normal Realism is a philosophy *involved* in science, being essentially, a resurgent natural philosophy.

Whereas Mach’s philosophy emphasised the dependence of science on ‘sense-data’, Normal Realism also emphasises, the part played by ‘language’ in the organising of *all* sense experience. In other words, in Normal Realism, ‘sense-data’ includes not only sense and instrument data

but also all the *informational* data transacted between observers in what is generally known as *language*.

In Normal Realism, what we *perceive* is never simply ‘sense-data’, but rather whole relational systems of the same. For example, we perceive that a barn door has shape, size, mass, position in relation to other objects, and so on, by observing, with all our sensations in concert, what happens when we push it or walk into it, or when it slams against its frame *etc.* All things which we perceive are, without exception, systems of interrelation which the object has with other objects. When the fullest possible account of all those interrelations is completed, nothing remains, behind and beyond perception, to which the classical term ‘thing-in-itself’ can be meaningfully applied.

In Normal Realism, then, there is no ‘God’s-eye-view’ presumption of intuiting the ‘absolute’ existence of entities underlying all our direct and instrumental perceptions of them. Scientific knowledge of the physical world rests, not on this Realism of classical Physics, but always on our very best logical interpretations of *phenomena*. These *phenomena* are not the subjective ‘pale shades’ of those ‘underlying realities’ that in Cartesian dualism are known directly ‘only to God’. They are the one and only physical reality which is manifest in all the *observational* (*i.e.*, sensory and instrumental) categories articulated by the barest, most economical use of communal language.

The main difference between this *Normal* language and the incorrigible ‘Ordinary Language’ of the ‘Linguistic’ philosophers is that the language of Normal Realism is not fixed and immutable, as ‘Ordinary Language’ was supposed to be. Language, in Normal Realism, is essentially developmental, evolving as our communal experience evolves, and hence, always, in principle, revisable in the face of natural uncertainty. As long as we remain careful in attending to its continued construction, reconstruction and maintenance, this evolving language is *objective* in a way that requires no reference to anything beyond itself. The ultimate aim of this empirical or developmental scientific language, is, by the process of free and continual logical elimination or error, to winnow out from contemporary scientific language, the true universal language of nature, which the Ancient Greeks called *logos*, from all the fanciful jargon and theoretical chaff. In short, the mantra of the Normal Realist might well be, ‘I have no

idea what Ultimate truth is; but by the careful use of logic, I sure enough know what isn't!

Throughout this process, an important principle to adhere to is the principle commonly known as 'Ockham's Razor'. This principle emphasises the need for conceptual economy in ideas which interpret observational information. This principle was first formally introduced into science in the fourteenth century by William of Ockham, but the idea can be traced back to the ancient Greeks, who showed a marked desire for simplicity of explanation. This also accords with Mach's positivistic criterion of the plainest and most unadorned descriptions of natural phenomena. However, it must be noted that sometimes, in applying Ockham's Razor, some of the predictive value of a theory may be lost. Take, for example, the paradigm shift from the Ptolemaic to the Copernican view of the Solar System. Despite its conceptual superiority, the Copernican System was much less exact, in observational records, than its predecessor. This means that for pragmatic purposes, a new conceptual paradigm and an old one which is incommensurable with it may have to exist, confusingly, side-by-side for as long as it takes for any conflict to settle.

5. Light

In applying its methods as much to Philosophy as to Physics, Normal Realism logically integrates the two. This it does by removing the last relics of the Cartesian separation of the **observer** from **physical reality**. Not least of these persistent relics is the classical 'speed of light' interpretation of the dimensional constant c in Einstein's Special Relativity, which separates the object observed from the observer of the object in the Cartesian way. Hence, the classical Cartesian separation has become what is currently called the 'Einstein separation', the logical fallacy of which is the same as in Descartes' separation of observational reality into objects as we perceive them and those same objects as perceived by 'God'. Thus, despite there being no possible way known to empirical science in which the light by which we see things can be observed or detected 'travelling in *vacuo*', we nevertheless assume that our ideas of what a thing is 'in itself' and our ideas of 'how we perceive it' are fundamentally separated, as in Cartesian dualism, by this metaphysical agency of light travelling in the way the second axiom of Special Relativity asserts.

In the new paradigm, the *light* that was historically regarded (and still is by most Physicists) as an incidental mediator of information about an inexorably remote ‘world beyond’ becomes *that very information itself*, which is the fount of all our knowledge of space, time, matter and motion. In this new way of thinking, we cease trying to match our human perceptions and conceptions to the world as ‘God’ might be presumed to perceive it, and settle for solving problems arising from *logically conflicting interpretations* of ordinary and instrumental experience. This prompts us to revise our ideas in the various categories of observation and conception towards satisfying the need for what is commonly known as *understanding*. For example, there are no ‘atoms’ we can ever know as ‘things in themselves’ as opposed to our *ideas* of them. Nor is there any self-sufficient space or time beyond what we can actually observe and measure. So far as Normal Realism is concerned, anything we may identify in *thought and language* as ‘atoms’, ‘electrons’, ‘charges’, ‘fields’, ‘photons’ or whatever, is revisable in the light of rational reflection on ongoing experience and observation.

6. Atomism versus Holism

If a *realistic* approach to Physics is taken, there is the question of whether what exists in any real sense is the whole of an object or only its constituent parts. To the ancient Greeks it seemed that only the ultimate bits of things were real and that collections of those bits were unreal in the sense that they could not exist without those component bits. But those ultimate bits (‘atoms’) are invisible, so those things we observe, which are not atoms, but collections of atoms, have no reality in themselves and therefore no powers of their own destinies. And since those invisible atomic bits were regarded as purely mechanical, acting individually in a ‘space’ all of their own, without any cognizance whatever of one another, far less of those wholes of which we say they are the parts, the ‘universe’ inasmuch as it made sense to speak of it, was regarded as a set of ultimate particles, whose movements and contacts were either purely accidental, or as in later theories mechanically preordained.

With our Normal Realist philosophy, however, that classical *reductionist* (*atomistic*) argument is *reversed*. Starting with things that we actually

observe, we say that the ultimate parts, or observational elements, into which those things may be sensibly or instrumentally analysed are simply *ideas* that have no qualities or characteristics other than in the contexts of those *observational wholes* of which they are the analysed parts.

But if the atomic events that are the parts of things are mechanically and geometrically unconnected, then what make those things hang together in the way they do? Our answer is, *the logical character of the whole*. For instance, to return to an example which I gave in an earlier lecture, the reason why the Moon circles Earth is simply because each body has a *holistic* relation, traditionally called *angular momentum* relative to the other body and to other bodies in general and because that angular momentum is a paired and overall conserved relation. In this way, the important thing, from our relationist point of view, is to identify the law relating to the whole. Once this holistic law is correctly determined, the motions of the parts are sufficiently explained.

In the next lecture, I will examine further the role of any observer and the role played by the constant c , classical labelled as 'the speed of light' in Physics.

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